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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
 B.Tech Degree S8 (R,S) Examinations April 2026 (2019 Scheme)

Course Code: EET426
Course Name: SPECIAL ELECTRIC MACHINES

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

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| 1 | Enumerate the applications of permanent magnet brushless DC motor. | (3) |
| 2 | What are the features of permanent magnets used in permanent magnet DC motors. | (3) |
| 3 | Describe the various types of windings employed in a stepper motor. | (3) |
| 4 | Draw and explain the dynamic characteristics of a stepper motor. | (3) |
| 5 | Explain the relevance of the rotor position sensor in the operation of a switched reluctance motor. | (3) |
| 6 | List any two causes for noise in a switched reluctance motor and explain how they can be mitigated. | (3) |
| 7 | Explain the role of servomotors in automation systems. | (3) |
| 8 | Enumerate the applications of the servomotor. | (3) |
| 9 | Draw the equivalent circuit and phasor diagram of a repulsion motor. | (3) |
| 10 | Explain the relevance of compensating winding in an AC series motor. Also draw its phasor diagram. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

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| 11 | With a neat sketch, explain the principle of operation of a permanent-magnet Brushless DC motor with 180 ° commutation. | (14) |
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OR

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| 12 | a) With a neat block diagram explain the sensor less control scheme of permanent magnet synchronous motor. | (7) |
| | b) With neat sketches, explain the construction and working of a permanent magnet synchronous motor. | (7) |

Module II

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| 13 | a) With neat sketches and truth tables, explain the different modes of operation of a four-phase six-pole single-stack variable reluctance stepper motor. | (14) |
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OR

- 14 a) Explain the construction and any one mode of operation of a multi stack variable reluctance motor. (10)
- b) Explain the microstepping operation in a hybrid stepper motor. (4)

Module III

- 15 a) With a neat circuit diagram, explain the working of any two power electronic converter circuits of a switched reluctance motor. (8)
- b) From fundamentals, derive the expression for the torque developed in the synchronous reluctance motor. (6)

OR

- 16 a) Explain with figures, the construction of any two rotor configurations of a synchronous reluctance motor. (8)
- b) With the help of a block diagram, explain any one control scheme of a switched reluctance motor. (6)

Module IV

- 17 a) With neat sketches, explain the construction of various types of rotors used for an AC servomotor. (9)
- b) Draw and explain the torque-speed characteristics of a DC servomotor. (5)

OR

- 18 a) Derive the expression for the transfer function of a field-controlled DC servomotor. Also, draw its block diagram. (8)
- b) With a neat figure, explain the construction and operation of a series-split-field DC servomotor. (6)

Module V

- 19 a) Derive the expression for thrust of linear force developed in a linear induction motor. Also enumerate any three advantages of linear motor over rotational motor. (8)
- b) Explain the transverse edge and end effects in a linear induction motor. (6)

OR

- 20 a) Explain the construction and principle of operation of a hysteresis motor. Sketch and explain the torque speed characteristics of a hysteresis motor. (10)
- b) Draw and explain the equivalent circuit and phasor diagram of a universal motor. (4)
